Ex. No: 5Inter Process Communication (IPC)

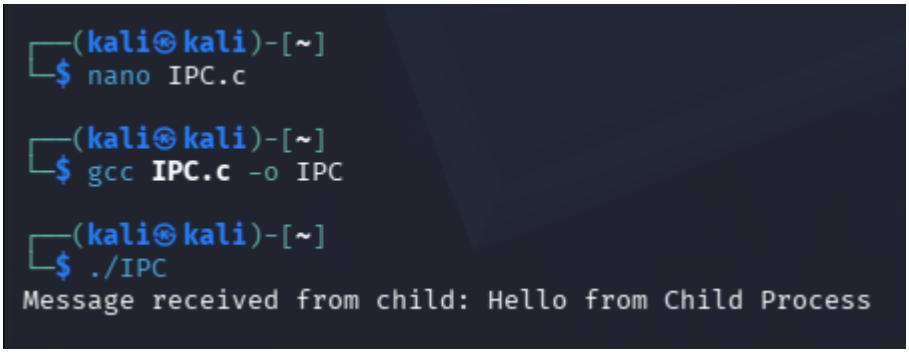
C PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

int main()
{
   int fd[2];
   pid_t pid;
   char message[] = "Hello from Child Process";
   char buffer[100];

   if (pipe(fd) == -1)
   {
      perror("Pipe failed");
      return 1;
   }

   pid = fork();

   if (pid < 0)
   {
      perror("Fork failed");
```

```
    return 1;
}

if (pid == 0)
{
    // Child process
    close(fd[0]);

    write(fd[1], message, strlen(message) + 1);

    close(fd[1]);
    exit(0);
}
else
{
    // Parent process
    wait(NULL);

    close(fd[1]);

    read(fd[0], buffer, sizeof(buffer));

    printf("Message received from child: %s\n", buffer);

    close(fd[0]);
}
```

```
    return 0;
}
```

Output:



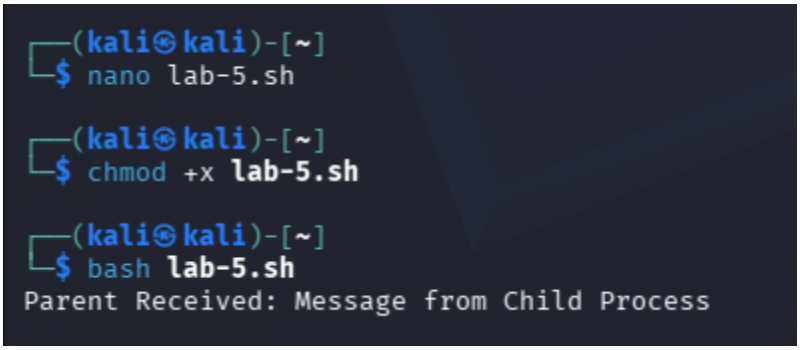
SHELL SCRIPT:

```
#!/bin/bash
echo "Hello from Child Process" | cat
```

Alternative Shell Script

```
#!/bin/bash
(
echo "Message from Child Process"
) | while read msg
do
echo "Parent Received: $msg"
done
```

Output:


┌──(kali㉿kali)-[~]
└─$ nano lab-5.sh
┌──(kali㉿kali)-[~]
└─$ chmod +x lab-5.sh
┌──(kali㉿kali)-[~]
└─$ bash lab-5.sh
Parent Received: Message from Child Process